Proposal of team software project

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Study program: Computer Science - Software and Data Engineering

Project title: Service Application for Recommender Systems with Sparse Autoencoders

Project type: Research project

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Expected end: 2026-04-31

1 Introduction

This research project is part of the GAČR initiative, specifically under the project LLM4MORS (GAČR 25-16785S) which is managed by the above-mentioned supervisor (Mgr. Ladislav Peška, Ph.D.) and consultants. The LLM4MORS project focuses on leveraging Large Language Models (LLM) in multi-objective recommender systems. The primary aim is to enhance the explanation[1] of the recommendation and facilitate user (or editor) steering[2] of recommendations. Currently, the LLM4MORS project intensively explores the use of Sparse Autoencoders (SAE) for creating interpretable representations of user profiles and associated steering mechanisms.

The need for this application arises from the complexity and variability in approaches to integrating, training, and evaluating SAE within recommender systems. This process is inherently multi-step, with each step offering numerous alternatives for implementation and evaluation. The existing work that utilizes SAE in recommender systems (as referenced in 1.1), which discusses completed such projects under LLM4MORS represents only one of many possible pathways, highlighting the necessity for a framework that can automate future research and enable more effective exploration of the problem space. This project aims to provide such a framework, moving beyond the current ad-hoc methods towards a systematic solution.

The research project focuses on implementing a service application designed to test, analyze, and evaluate recommender systems utilizing SAE. The application will support the development of these systems by providing insights into their performance and interpretability in a an automated manner, enabling continuous testing and evaluation of model variants.

[1]The information about why the item was recommended by the complex model

[2]Guiding or influencing recommendation outcomes

1.1 Relation to other projects

This research project partly builds on previous work within the LLM4MORS project, particularly leveraging the integration of Sparse Autoencoders (SAE) in recommendation contexts. It is inspired by:

- The group's manuscript "From Knots to Knobs: Disentangling Collaborative Filtering Autoencoders with Sparse Autoencoders" currently submitted to WWW 2026, where the concept of connecting SAE and recommender systems was first introduced.
- The research paper "SAGEA: Sparse Autoencoder-based Group Embeddings Aggregation for Fairness-Preserving Group Recommendations" by Bc. Vít Koštejn, Mgr. Ladislav Peška, Ph.D. and Mgr. Martin Spišák which applied SAE in a specialized type of recommender system.

This project aims to build on these foundations by developing a robust framework for systematic testing, analysis, and evaluation of SAE-based models. The goal is to streamline the exploration of various implementation and evaluation pathways, transitioning from ad-hoc methods to a structured approach that enhances research effectiveness.

2 Project description

The application will implement a framework for executing multi-step pipelines (see Figure 1), which users can utilize for both development and analysis of recommender systems with SAEs. These pipelines will be constructed using plugins that implement a defined API, with the design of this API being a central part of the project. The researcher, acting as the user, will be able to implement methods within these plugins, with some default implementations provided. This approach allows for flexible experimentation and validation of new methods, especially in unexplored areas like neuron labeling and evaluation.

The solution will not only involve the formalization of the pipeline/framework (plugins APIs) but also the implementation of several alternatives for individual steps within the pipeline. This includes, for example, varying methods for dataset interaction, collaborative filtering algorithms, and neuron labeling techniques.

While the user will be able to access core functionalities via a command-line utility, the primary interface will be provided as a web application. This application will enable users to create, manage, and execute pipelines composed of plugins. It will also facilitate the storage, analysis, and visualization of pipeline results, ensuring reproducibility and comprehensive analysis.

The goal of the project is to prepare an experimental environment for the development and validation of pipelines. This will enable further exploration in this research domain by the research group and particularly by the solver, who plans to base his master's thesis on this project. The thesis will focus on methods for neuron interpretability and labeling, leveraging the framework as a practical tool.

The project will be primarily implemented in Python, with FastAPI employed as the web server framework. MLflow will be utilized for model registry and artifact storage, providing a structured approach to managing model versions and their associated data. Additional tools for web development, visualization, and data management will be selected at a later stage.

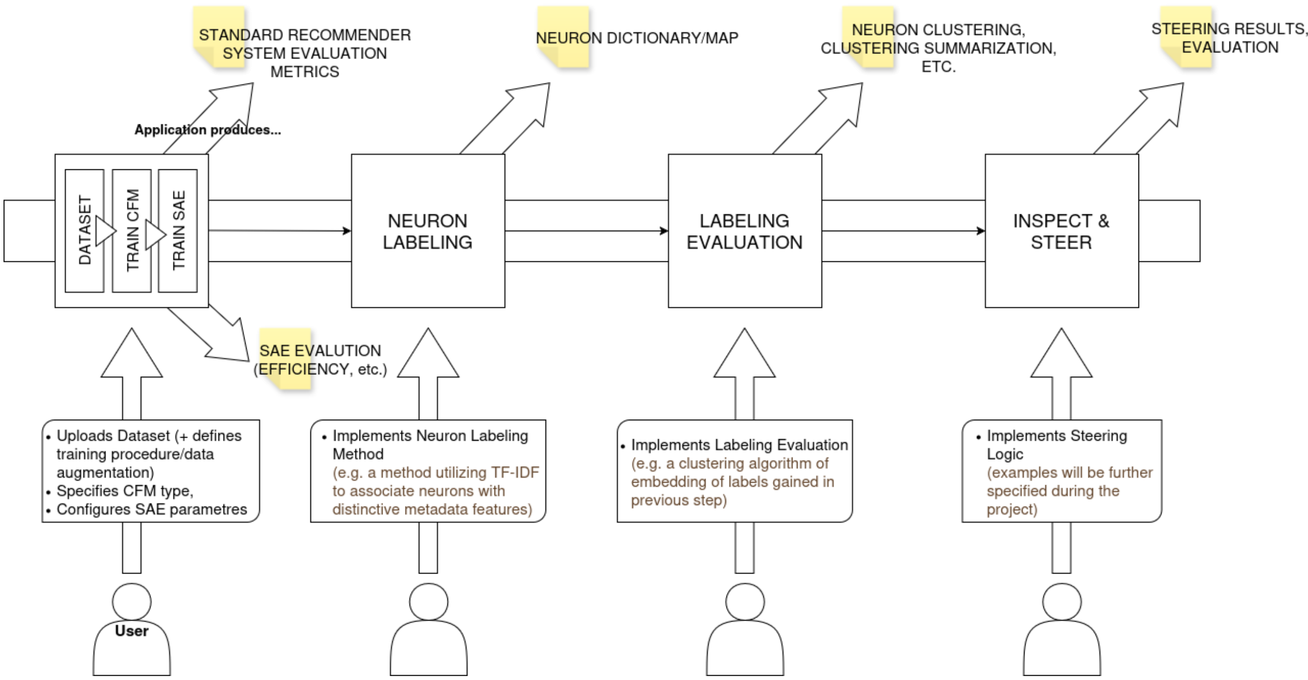

Figure 1: Diagram of the main application flow (alias training pipeline). In the first step (1), the training of the Collaborative Filtering Model (CFM) — a model used for recommendation tasks — is performed using the provided dataset, followed by the training of the embedded Sparse Autoencoder (SAE). This step produces standard evaluation metrics for recommender models, along with the performance measures of the SAE. Subsequently, in step (2), neuron labeling takes place — that is, interpreting individual neurons based on the model's behavior during validation and the characteristics of data samples. In the next step (3), the evaluation of the obtained labeling is performed, which can be realized, for instance, through neuron clustering or other analytical techniques. Finally, the pipeline concludes with the steering phase (4), where insights gained from previous steps are used to modify the recommendation process — for example, by strengthening neurons associated with certain concepts according to their labels — and to observe the resulting impact.